

Yiquan Wang

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Education

National Base for Mathematical Research and Teaching Talents, Xinjiang University <i>B.S. in Mathematics and Applied Mathematics</i>	Urumqi, Xinjiang 2023.09 – 2027.06
Tsien Excellence in Engineering Program, Tsinghua University & X-Institute <i>Joint Program, X Scholar</i>	Shenzhen, Guangdong 2024.06 – 2027.06
Institute of Neurological and Psychiatric Disorders, Shenzhen Bay Laboratory <i>Visiting Student, Wen Yuan's Research Group</i>	Shenzhen, Guangdong 2025.07 – 2025.09

Selected Publications

Direction 1: Representation Learning & Biomolecular Mechanisms

- [1] **Wang, Y.**, Cai, M., Dong, Y., et al. (2025). From Signal to Symphony: Exploring 2D Sequence Representations for Protein Function Prediction. *Journal of Chemical Information and Modeling* (JCR Q1, CAS Q2 Top).
- [2] **Wang, Y.***, Cai, M., Ma, Y., & Wei, K. (2026). Sequence-context-aware decoding enables robust reconstruction of protein dynamics from crystallographic B-factors. *Learning Meaningful Representations of Life (LMRL) Workshop at ICLR 2026*.
- [3] **Wang, Y.**, Ma, Y., Chang, Y., et al. (2025). Diffusion Models at the Drug Discovery Frontier: A Review on Generating Small Molecules versus Therapeutic Peptides. *Biology* (JCR Q1, CAS Q2).
- [4] Wang, X.†*, **Wang, Y.†***, Huang, T. Y., et al. (2025). Octopus Inspired Optimization (OIO): A Hierarchical Framework for Navigating Protein Fitness Landscapes. *In 2025 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*. IEEE. (CCF-B, Co-first & co-corresponding author).
- [5] **Wang, Y.***, & Huang, T. Y.* (2026). Comment on: "GLM7 – A Novel Composite Glycolipid Index Derived from Routine Health Indicators for Enhanced Diagnosis and Prediction of Multimorbidity". *Advanced Science*. (correspondence, JCR Q1, CAS Q1, First & corresponding author).
- [6] **Wang, Y.***, Cai, M., & Huang, T. Y. (2025). AI for disease prediction: Performance insights and key limitations. *Journal of Clinical Neuroscience*, 138, 111360. (correspondence, JCR Q3, CAS Q4, First & corresponding author)

Direction 2: First-Principles of Biological Complexity Based on Mathematical Physics

- [7] **Wang, Y.*** (2026). Spectral entropy of the discrete Hasimoto effective potential exposes sub-residue geometric transitions in protein secondary structure. *arXiv preprint arXiv:2602.21787*. (Sole & corresponding author).
- [8] **Wang, Y.*** (2026). Piecewise integrability of the discrete Hasimoto map for analytic prediction and design of helical peptides. *arXiv preprint arXiv:2602.16255*. (Sole & corresponding author).
- [9] **Wang, Y.*** (2026). Structural barriers of the discrete Hasimoto map applied to protein backbone geometry. *arXiv preprint arXiv:2602.13160*. (Sole & corresponding author).

Direction 3: Interdisciplinary Explorations & Collaborative Synergies

- [10] Wang, X., **Wang, Y.**, & Huang, T. Y. (2025). Crypto-ncRNA: Non-coding RNA (ncRNA) Based Encryption Algorithm. *ICLR 2025 Workshop on AI for Nucleic Acids*. (Co-first author).
- [11] **Wang, Y.***, Chang, Y., Zhang, J., Cai, M., Yan, J., Zhao, H.*, & Alzheimer's Disease Neuroimaging Initiative. (2026). Geometric Fragility in Alzheimer's Disease: Probing the Loss of Hippocampal Hierarchical Abstraction via Contrastive Point Cloud Modeling. *Proceedings of the Annual Meeting of the Cognitive Science Society (CogSci 2026)*; also presented at *LMRL Workshop at ICLR 2026*. (First & corresponding author).
- [12] **Wang, Y.**, Liu, Z., Zai, J., Chen, J., & Sabir, E. (2026). Resilient AI Infrastructure by Design: A Spatially-Aware Framework for Tolerating Clustered Failures. *In 4th Annual AAAI Workshop on AI to Accelerate Science and Engineering (AI2ASE)*.

- [13] **Wang, Y.***, Huang, T. Y., Gao, Q., Chang, Y., & Zhang, J. (2026). AI Driven Discovery of Bio Ecological Mediation in Cascading Heatwave Risks. *ICLR 2026 Workshop on Foundation Models for Science: Real-World Impact and Science-First Design (FM4Science)*. (First & corresponding author).
- [14] **Wang, Y.***, Cai, M., Zhang, J., Chang, Y., & Yan, J. (2026). Adaptive Decision-Making in Multi-Stage Production: A Framework for Cost Optimization under Sampling Uncertainty. *Applied Operations and Analytics*.
- [15] Zhao, H., Shi, W., Liu, C., & **Wang, Y.** (2025). Boundary Box-Guided Targeted Adversarial Attacks with Semantic Perturbation. *In 2025 IEEE International Conference on Systems, Man, and Cybernetics (SMC)* (pp. 1604-1609). IEEE. (CCF-C, EI).
- [16] Zuo, J., **Wang, Y.**, & Yibulayin, X. (2026). Relative-to-human Benchmark Cognitive Divergence and Semantic Comprehensibility in Chinese–Uyghur LLM Translation. *Frontiers in Psychology*.
- [17] **Wang, Y.***, Zhang, J., & Chang, Y. (2024). A probability prediction model for flood disasters based on Multi-layer Perceptron. *Journal of Physics: Conference Series* (Vol. 2905, No. 1, p. 012003). IOP Publishing.

Research Projects

Tsinghua University Tsien Excellence in Engineering Program ESRT

2024.08-2025.10

- **Project:** From Signal to Symphony: Exploring 2D Sequence Representations for Protein Function Prediction.
- Predicting protein function from its primary sequence is a fundamental challenge in computational biology. While deep learning has excelled, the optimal representation of sequence data remains an open question. This study explores protein sonification—the conversion of amino acid sequences into 2D spectrograms—as a representation for this task. To facilitate this investigation, we developed a benchmark dataset of 18,000 sequences spanning 12 functionally diverse protein classes. Our systematic evaluation suggests that the structural transformation from a 1D sequence to a 2D spectrogram may be a key contributor to the model's predictive performance. This observation is supported by ablation studies where models using either purely visual or acoustic features from the spectrogram demonstrated effective standalone performance, suggesting that the representation itself is a key source of this capability. For instance, a model using a sonification map without explicit biophysical meaning achieved 81.08% accuracy, while our biophysically-informed model reached 84.00%, indicating that such domain knowledge may offer a modest performance benefit. When trained from scratch on our dataset, our fusion model achieved performance comparable to or slightly exceeding that of standard transformer architectures like ESM-2 and ProtBERT, suggesting its potential for data efficiency in this specific context. The model's potential for generalizability was further supported by its performance on the external CARE enzyme classification benchmark, where it achieved 90.44% accuracy. Finally, as a proof-of-concept, we explore the utility of our encoding to guide a diffusion model in generating novel GFP variants, which were assessed for structural viability using computational methods. Our work provides evidence suggesting that the utility of sonification in this context may stem largely from its representational structure, offering a perspective on feature engineering for biological sequences.
- [Project Repository](#). Supervised by Prof. [Kai Wei](#).

National Undergraduate Innovation Training Program

2025.04 – 2026.04

- **Project:** Copy Number Variation Conditional Diffusion Model: For Alzheimer's Disease Risk Assessment.
- This project aims to integrate copy number variation (CNV) features from whole-genome sequencing data with multi-dimensional clinical data such as metabolic indicators. Leveraging the success of diffusion models in processing high-dimensional simulation data and protein phenotype prediction, we will construct a comprehensive framework consisting of CNV feature encoding, genomic region attention, and conditional U-Net diffusion modules. This will simulate CNV distribution changes and evolutionary processes in the genome, analyze the specific role of CNV in regulating Alzheimer's disease pathways, and ultimately improve disease risk assessment and early intervention accuracy.
- Supervised by Prof. [Kai Wei](#).

Chinese Academy of Sciences (CAS) Innovation Practice Training Program

2024.11 – 2025.09

- **Project:** A Knowledge Graph-based Q&A System for Heatwave Disaster Adaptation.
- Compound heatwaves increasingly trigger complex cascading failures that propagate through interconnected physical and human systems, yet the fragmentation of disciplinary knowledge hinders the comprehensive mapping of these systemic risk topologies. This study introduces the Heatwave Discovery Agent HeDA as an autonomous scientific synthesis framework designed to bridge cognitive gaps by constructing a high fidelity knowledge graph from 8,111 academic publications. By structuring 70,297 evidence nodes, the system exhibits enhanced inferential fidelity in capturing long tail risk mechanisms and achieves a significant accuracy margin compared to standard foundation models including GPT 5.2 and Claude Sonnet

4.5 in complex reasoning tasks. The resulting topological analysis reveals a critical bio ecological mediation effect where biological systems function as the primary non linear amplifiers of thermal stress that transform physical meteorological hazards into systemic socioeconomic losses. We further identify latent functional couplings between theoretically distinct sectors such as the heat induced synchronization of power grid failures and emergency medical capacity saturation. These findings elucidate the dynamics of compound climate risks and provide an empirical basis for shifting adaptation strategies from static sectoral defense to dynamic cross system resilience.

- [Project Repository](#). Supervised by Researcher [Yong Ge](#).

Provincial Undergraduate Innovation Training Program

2024.03 – 2025.06

- **Project:** Research on the Generation Cyclability of Cartesian Product Graphs and Related Problems.
- The k -ary n -cube (Q_n^k) is a critical topology for large-scale computing systems powering modern AI and HPC workloads, where fault tolerance is paramount. Traditional fault models, which assume faults are independent and random, yield unrealistic resilience estimates because real-world failures are spatially correlated, manifesting as topological clusters. This paper introduces the Region-Based Fault (RBF) model, a new paradigm that addresses this gap by directly modeling this spatial correlation. Our primary contribution is a proof that for odd $k \geq 3$ and $n \geq 2$, the Q_n^k remains Hamiltonian-connected—a property vital for deadlock-free routing and efficient task scheduling—under a sufficient set of RBF conditions. We present a constructive algorithm that finds a Hamiltonian path by leveraging an adaptive decomposition strategy. Experimental analysis demonstrates that our approach significantly enhances fault tolerance and remains robust far beyond its conservative theoretical guarantees. This work provides a practical, high-performance solution for maintaining connectivity in systems where fault clustering is prevalent.
- [Project Repository](#). Supervised by Prof. [Eminjan Sabir](#).

Tsinghua University Tsien Excellence in Engineering Program ORIC

2025.12 – 2026.12

- **Project:** Application of Artificial Intelligence in Whole Genome Selection Breeding.

Chinese Academy of Sciences Innovation Practice Training Program

2025.11 – 2026.11

- **Project:** Urban Big Data, Multimodal Fusion and Spatial Intelligence Analysis (Second Author).

Urumqi Sandaoban Town Cultural Exhibition Project

2026.01 – 2026.06

- **Project:** Forging a Strong Sense of Community for the Chinese Nation Exhibition Project (Participant).

Professional Experience

Beijing Frontier Research Center for Biological Structure, Tsinghua University

Beijing, China
Intern2025.07 – Present

- Developed the [Peptide Design Competition Website](#), the [Polypeptide Structure Database](#) and engaged in protein and polypeptide design research. Supervised by Prof. Yafei Yuan and [Prof. Tong Wang](#).

Institute of Software, CAS & Huawei Mindspore Community

Remote
Intern2024.09 – 2025.03

- Implemented a VGG19-based model for Pollock-style art generation, focusing on fractal and turbulent feature extraction and the creation of NFT-based digital labels. Featured on [Huawei's official Wechat](#).

Awards & Honors

- **Smart Chemical City Challenge (Xinjiang Station):** Second Prize2025
- **Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM):** National Second Prize2025
- **SynBio Challenges:** Silver Award × 22025
- **2025 X-Fusion "Global Innovator Fusion Conference":** Best Poster Award2025
- **The Mathematical Contest in Modeling (MCM):** Honorable Mention2025
- **Alibaba Cloud Tianchi University Student Competition:** National Finals, 17th Place2024
- **APMCM Asia-Pacific Mathematical Modeling Competition:** National Third Prize2024
- **“Alpha Egg Cup” National Go Championship:** 15th Place2024

- **Xinjiang Youth Amateur Go Competition:** 53rd Place 2024
- **Hunan Province Spring Cup Go Competition:** 7th Place 2024
- **National Youth Intellectual Sports Meeting Go Competition:** 9th Place 2024
- **Xinjiang University Vulnerability Reporting Honor** 2023
- **“Tianshan Fixed Network Cup” Cybersecurity Skills Competition:** 7th Place, Xinjiang Region 2023

Academic Activities

Peer Reviewer:

- ICLR 2026 Workshop LMRL
- ICLR 2026 Workshop FM4Science
- ICML 2026 AI4Science Workshop
- NeurIPS 2025 AI for Science Workshop
- NeurIPS 2025 MATH-AI Workshop
- ICLR 2025 Workshop on AI for Nucleic Acids
- ICLR 2025 Workshop on GenAI Watermarking
- ICML 2025 Workshop on AI for Math
- Urban Climate
- Mini-Reviews in Medicinal Chemistry
- Current Science
- F1000 Research

Academic Engagement:

- Tsinghua University-Peking University Center for Life Sciences Summer Camp 2025.07
- Shenzhen Bay Laboratory / Shenzhen Medical Academy of Research and Translation Summer Research 2025.07 – 2025.09
- Tsinghua University Tsien Excellence in Engineering Program – Zero One Scholar 2024.06 – 2027.06
- AI Winter School, Brown University Department of Physics 2025.01
- CAAI Artificial Intelligence and Technology Ethics Training Course 2024.09 – 2024.12
- Fudan University Summer School of Mathematical Logic 2024.08
- Jinan University Guangdong Thousand Villages Survey Project 2024.08
- Wuhan University National Tianyuan Mathematics Center Discussion Class 2024.03 – 2024.06

Website Development

- **Peptide Design Competition, Tsinghua University:** <https://www.fbs.frcbs.tsinghua.edu.cn/competition/2025Peptide/>
- **Polypeptide Structure Database, Tsinghua University:** <https://www.frcbs.tsinghua.edu.cn/cpdb/>
- **Tong Wang (Tsinghua University) Research Group:** <https://tongwang.vercel.app/>

Skills & Interests

Technical Skills:Python, C/C++, MATLAB, LaTeX, Linux, HTML/CSS/JavaScript, Single-cell Data Analysis, Multi-omics (Genomics, Proteomics, Transcriptomics), Homology Modeling, Molecular Dynamics Simulation

Research Interests:Artificial Intelligence, Deep Learning, AI for Science, Bioinformatics, Computational Biology, Representation Learning, Protein Folding, Mathematical Physics (Nonlinear Schrödinger Equation & Protein Folding), Mathematical Modeling, Neuroscience

Languages: Chinese (Native), English (Professional Working Proficiency)